

The big idea. A **function** gives exactly **one output** for each input. Everything else (domain, range, where it rises, how fast it changes, what you get when you combine or undo it) is a question about those inputs and outputs. Ask: *is it x or is it y that I am being asked for?*

ANALYZING A FUNCTION, STEP BY STEP

- 1

Is it a function? No input may have two outputs. On a graph, use the **vertical line test**.
- 2

Domain = allowed inputs (x -values). Look for division by 0, even roots of negatives, logs of non-positives.
- 3

Range = outputs (y -values). On a graph, read how low and how high the curve goes.
- 4

Increasing or decreasing? Read left to right, and answer with x -intervals, never y -values.
- 5

Peaks and valleys are local max and min. Name the point, or the y -value if asked for the value.
- 6

Combine or undo. Composition works inside-out. An inverse swaps input and output.

Most common mistakes: (1) Giving the y -values when asked for the domain (or the reverse). (2) Reading increasing/decreasing as y -intervals. (3) Evaluating $(f \circ g)(x)$ in the wrong order: g **goes first**. (4) Reading $f^{-1}(x)$ as $\frac{1}{f(x)}$. It is the **inverse**, not the reciprocal. (5) Using a closed bracket at an excluded value.

WHAT IS THE QUESTION ASKING?

THE QUESTION SAYS	IT MEANS	DO THIS
Domain of f	Allowed x	Rule out division by 0 and bad roots. Write in interval notation.
Range of f	Possible y	On the graph, lowest to highest height.
Is f one-to-one?	No repeated outputs	Horizontal line test: no line hits twice.
Find $(f \circ g)(2)$	$f(g(2))$	Compute $g(2)$ first, then put it into f.
Find $f^{-1}(7)$	Which x gives 7?	Solve $f(x) = 7$ for x.

FORMULA BANK

FUNCTION & ONE-TO-ONE Function: each input has one output (passes the vertical line test). One-to-one: each output has one input (passes the horizontal line test). Only one-to-one functions have inverses.	DOMAIN RULES Fractions: exclude where the denominator is 0. $\sqrt{g(x)}$: need $g(x) \geq 0$. $\ln g(x)$: need $g(x) > 0$. Polynomials: all real numbers.	INCREASING, DECREASING Increasing: graph goes up as you move right. Decreasing: goes down. Local max / min: highest or lowest point nearby (the turning points). Use parentheses for the x-intervals.
AVERAGE RATE OF CHANGE From a to b $\frac{f(b) - f(a)}{b - a}$ It is the slope of the line (secant) through the two points. Units: output per input.	COMPOSITION & INVERSE Composition $(f \circ g)(x) = f(g(x))$ Inverse $f(a) = b$ means $f^{-1}(b) = a$ To find a formula for f^{-1}: write $y = f(x)$, swap x and y, solve for y.	PIECEWISE FUNCTIONS Check which condition the input satisfies, then use only that rule. At a boundary, look at which piece has the $\leq$ or $\geq$.

Eight problems that cover the exam

Ask first: am I given an input, or an output?

Walkthrough 1: Function? One-to-one?

REPEATED VALUES

Table: $x = 1, 2, 3, 4$ and $y = 5, 7, 5, 9$. Is it a function? Is it one-to-one?

- 1 FUNCTION?** Each x appears once, so each input has one output: **yes**.
- 2 1-TO-1?** The output 5 comes from both $x = 1$ and $x = 3$.
- 3 ANSWER** A function, but **not** one-to-one (it fails the horizontal line test).

Walkthrough 2: Find the domain

RULE OUT BAD INPUTS

$$f(x) = \frac{\sqrt{x-3}}{x-5}$$

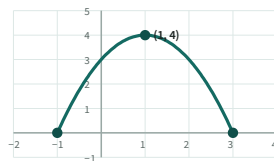
- 1 ROOT** $x - 3 \geq 0$, so $x \geq 3$.
- 2 BOTTOM** $x - 5 \neq 0$, so $x \neq 5$.
- 3 COMBINE** Start at 3 (included) and skip 5: $[3, 5) \cup (5, \infty)$

Walkthrough 3: Domain and range from a graph

READ X THEN Y

Find the domain and range of the graph shown.

- 1 DOMAIN** Leftmost point $x = -1$, rightmost $x = 3$, both solid: $[-1, 3]$
- 2 RANGE** Lowest height 0, highest 4 (the peak): $[0, 4]$



Solid endpoints: use brackets

Walkthrough 6: Piecewise function

USE THE RIGHT PIECE

$$f(x) = \begin{cases} x + 2, & x < 1 \\ x^2 - 1, & x \geq 1 \end{cases}. \text{ Find } f(-3), f(1), f(3).$$

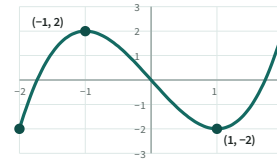
- 1 F(-3)** $-3 < 1$: top rule. $-3 + 2 = -1$
- 2 F(1)** $1 \geq 1$: bottom rule. $1 - 1 = 0$
- 3 F(3)** $3 \geq 1$: bottom rule. $9 - 1 = 8$

Walkthrough 4: Increasing, decreasing, extrema

INTERVALS IN X

Describe the graph shown on $[-2, 2]$.

- 1 INCREASING** It rises from the left end up to the peak at $x = -1$, then again after $x = 1$: $(-2, -1)$ and $(1, 2)$
- 2 DECREASING** Falls between the peak and the valley: $(-1, 1)$
- 3 EXTREMA** Local max at $(-1, 2)$. Local min at $(1, -2)$.



Turning points split the intervals

Walkthrough 7: Composition

INSIDE FIRST

$f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find $(f \circ g)(2)$ and a formula for $(g \circ f)(x)$.

- 1 G(2)** $g(2) = 4 - 3 = 1$ (inner function first).
- 2 THEN F** $f(1) = 2(1) + 1 = 3$
- 3 G OF** Put f into g : $(2x + 1)^2 - 3 = 4x^2 + 4x - 2$
- 4 ORDER** $f \circ g$ and $g \circ f$ are usually **different**.

Walkthrough 8: Inverse values and formulas

SWAP INPUT AND OUTPUT

$f(x) = 3x - 5$. Find $f^{-1}(10)$ and a formula for $f^{-1}(x)$.

- 1 F⁻¹(10)** Which input gives 10? Solve $3x - 5 = 10$: $x = 5$.
- 2 CHECK** $f(5) = 10$, so $f^{-1}(10) = 5$ ✓
- 3 FORMULA** Write $y = 3x - 5$ and swap: $x = 3y - 5$.
- 4 SOLVE Y** $y = \frac{x+5}{3}$, so $f^{-1}(x) = \frac{x+5}{3}$

Walkthrough 5: Average rate of change

SLOPE OF THE SECANT

$f(x) = x^2 - 2x$ from $x = 1$ to $x = 4$.

- 1 OUTPUTS** $f(1) = 1 - 2 = -1$ and $f(4) = 16 - 8 = 8$.
- 2 RATE** $\frac{8 - (-1)}{4 - 1} = \frac{9}{3} = 3$
- 3 MEANING** On average, f rises 3 units for each 1 unit of x .